

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Ratings

Imputed by category median

\$50.76

Avg. Purchase

Demographics, purchase details, and shopping behavior across 4 product categories.

Data Preparation in Python

01

Load & Explore

`pandas · df.info() · describe()`

02

Clean & Standardize

Impute missing ratings, snake_case columns

03

Feature Engineer

`age_group · purchase_frequency_days`

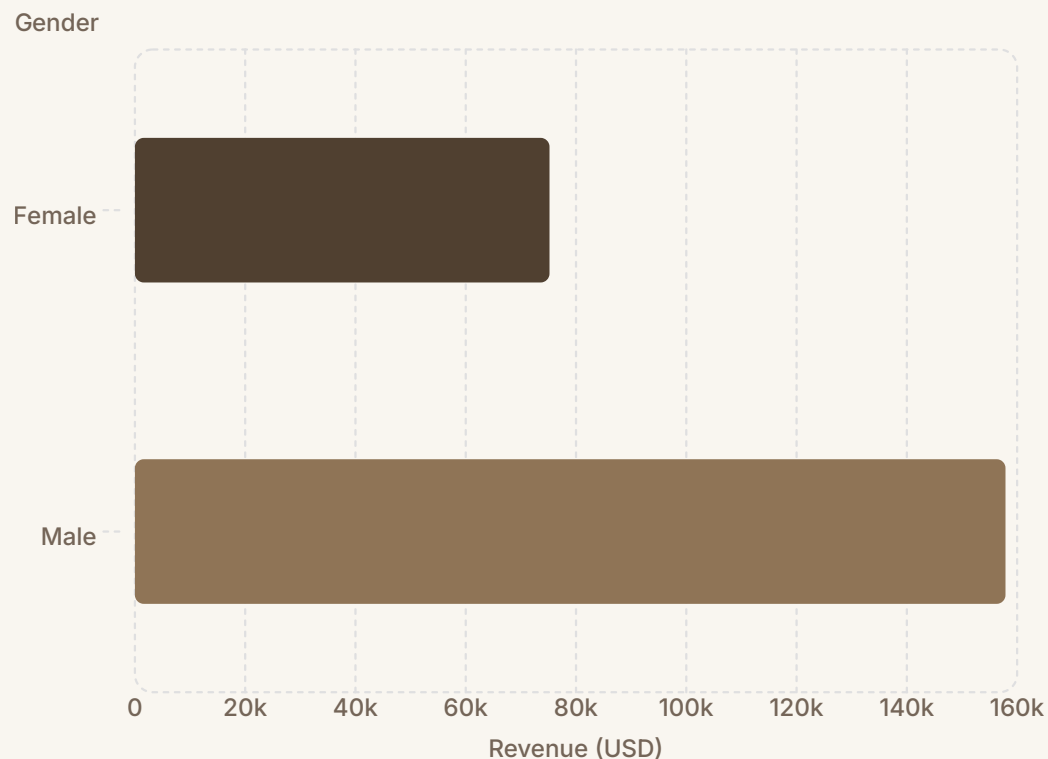
04

Load to PostgreSQL

Cleaned DataFrame → SQL analysis



Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers — \$157,890 vs. \$75,191.

- 📄 2,652 male customers vs. 1,248 female customers in the dataset.

Products & Shipping Insights

Top 5 Products by Rating

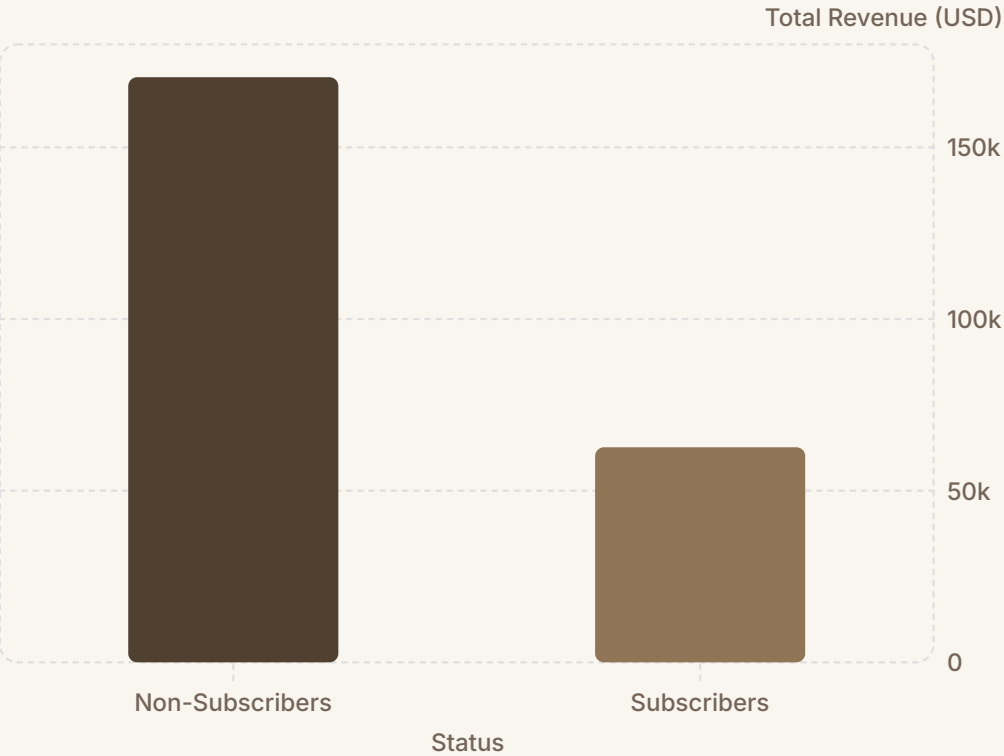
Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Avg. Purchase by Shipping

Shipping	Avg. Spend
Standard	\$58.46
Express	\$60.48

Express shippers spend **\$2 more** on average.

Subscribers vs. Non-Subscribers



Subscriber Profile

Subscribers spend nearly the same per transaction (\$59.49 vs. \$59.87) but represent only 27% of customers.

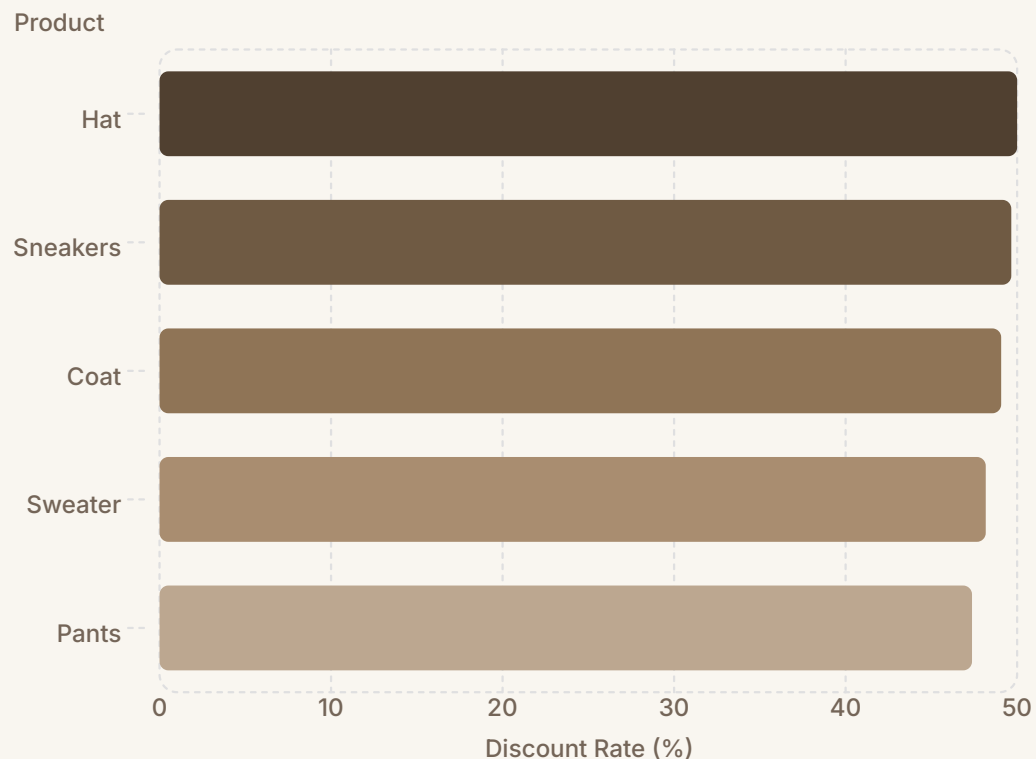
1,053

Subscribers

2,847

Non-Subscribers

Discount-Dependent Products



Key Finding

Top 5 products all have ~**47–50% discount rates** — nearly half of all purchases rely on promotions.



Heavy discount dependency may erode margins.
Review pricing strategy for these items.



Customer Segmentation

Loyal

3,116 customers — highest purchase frequency

Returning

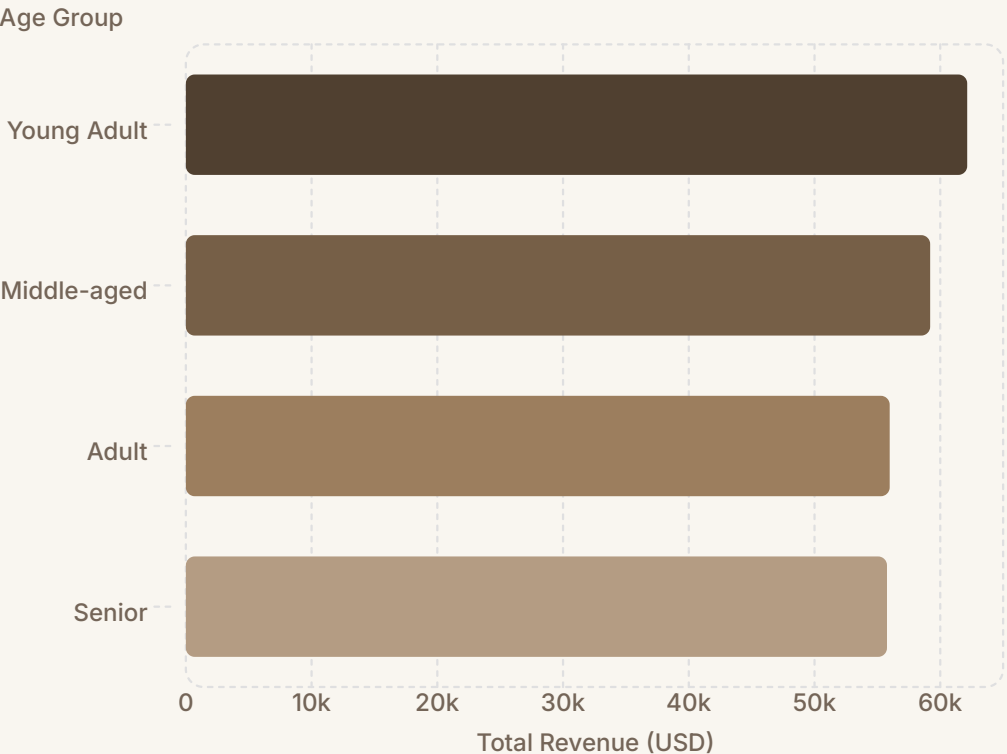
701 customers — moderate repeat behavior

New

83 customers — first-time buyers

958 repeat buyers (>5 purchases) are subscribers — loyalty drives subscription uptake.

Revenue by Age Group



Age Group Revenue

Young Adults lead with **\$62,143** in revenue. All four age groups contribute relatively evenly — no single segment dominates.

Average Age: 44

Range: 18–70 years

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers

